

PRATHEEK DHANANJAYA

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EDUCATION:

University of Illinois, Chicago
Masters in Computer Science

Chicago, Illinois
Aug 2024 – Present

- Coursework: Introduction to Networking, Network Security, Building Secure Computer Systems.

SJB Institute of Technology, Visvesvaraya Technological University
Bachelor of Engineering in Information Science and Engineering

Bengaluru, India
Aug 2018 – Aug 2022

- CGPA: 8.65/10, First Class with Distinction.
 - Coursework: Data Structures and Applications, Computer Networks and Security, Advanced Java and J2EE, Cryptography.
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SKILLS:

Programming Languages: Java, Maven, C, C++, Dart (used in Flutter), Python.

GUI and Databases: HTML, CSS; MySQL, MongoDB.

Tools and Platforms: Spring Tool Suite, MongoDB Compass, AWS CloudWatch and CloudFront, Git, Jenkins, Wireshark and Ollama.

EXPERIENCE:

Bosch Global Software Technologies (BSGW)
Software Engineer

Bengaluru, India
Jan 2023 – Aug 2024

- Provided **connected vehicle solutions** for Mahindra & Mahindra's Commercial Vehicle Segment.
 - Developed **REST APIs in Java 8** to support 'iMaxx' mobile and portal application, a product which allows owners and drivers of Mahindra vehicles to access live data, using **AWS S3 & SNS, MongoDB, MySQL**, etc. improving the user's experience through real-time notifications.
 - Some of the features I've developed are as follows:
 - Vehicle Checklist – Checklist that the owner publishes to every driver, which needs to be completed before every consignment.
 - Roadside Assistance – Providing an emergency SOS from the application and routing a message to the emergence contacts.
 - mShopee – nearby spare parts locator.
 - Fleet Announcement – Owner can create an announcement, create a poll for answers and send it to either drivers or fleet managers or both.
 - In-App Rating & In-App Language – User is prompted with a notification to rate the app after certain actions and is allowed to change the language to their preference from the option given. This feature introduced users by ~30% because of the regional language flexibility.
 - Subscription Package Management – OEM system admin can create and modify subscription packages to which the owners subscribe and employ among the drivers and managers.
 - Vehicle Reports – Generates individual and cumulative reports on vehicle alerts, vehicle expenses, etc, which had ~80% faster response time in generating it because of the incorporation of thymeleaf and **asynchronous framework**.
 - Multi Factor Authentication – Provides additional layer of security for data using Google Authenticator for OEM admin users in web portal.
 - In the process of development, made use of many AWS services like **CloudWatch** for tracking the logs, **IotCore** for publishing a payload to a subscribed channel, **Code Pipeline** to ensure proper deployment over **Jenkins** build and **AWS Lambda**.
 - While deploying, harnessed Jenkins pipeline to maintain a serverless deployment, fastened the deployment process by ~20%.
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INTERNSHIP:

CloudDepend Systems
Developer Intern

Bengaluru, India
Dec 2021 – Jun 2022

- Worked on projects using procedural language and flutter, to provide **interoperability between Android and iOS**.
 - 'Readability Analyzer,' a mobile application on both Android and iOS platforms, which achieved over 85% improvement in text readability scores through advanced analysis methods like Flesch-Kincaid, etc. using the **Dart** programming language and secured with **Google OAuth**.
 - 'xPeak,' an application which provides an integrated test client management solution to facilitate interoperation of real-world devices with equipment's under test, was developed in **C programming language**.
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ACADEMIC PROJECTS:

ProcProtect: LLM powered Malware Detection using /proc file system

- Developed a local **Linux cybersecurity daemon tool** leveraging the large language model Llama 3.2 (Ollama) for **on-device threat analysis**.
- Implemented **real-time /proc file system monitoring to detect malicious processes based on file path heuristics**.
- Enhanced system security by performing local analysis, ensuring data privacy and minimizing reliance on external servers.

Two Factor Authentication Using Morse Code

- Implemented an additional layer of security over the traditional PIN based authentication system, using **computer vision framework** and **Haar Cascade Algorithm**, through a keyless entry of passcode.
- This **python** model requires the user to enter a valid passcode through eye blinks, which is captured by the computer's camera module through **facial landmark detection**, in the form Morse Code, which had a success rate over 95% per authentication.
- This system helps in avoiding **Keylogger attacks** and increasing the security measures in places like ATM and Net Banking.

TO-DO List Mobile Application

- An application, developed in android studio, primarily used to organize and manage daily tasks, using **systrace**.
- Application developed, using **shared preferences**, also serves as a supervision tool for parents, where the parent assigns task to do to their children. The task can be flagged completed only by the parent, after the child uploads a photo to that note.

Event Management System

- A **console application** which purely runs on C++, harnessing the techniques of dynamic and linear hashing to store data in the form files.
- Application provided both a user side usage and an admin side usage, having access to perform **CRUD operations** on the events to be hosted.

Airline Ticket Reservation System with Currency Exchange (Personal Project)

- The application was developed in **Java**, where a user can create an account for login credentials, using which the user can book domestic or international flight tickets.
 - Implemented all the fundamental concepts of Java, along with mapping to a database. This application also allowed the user to exchange currency for the respective destination, if the user is booking an international ticket and provided the user with a compiled report.
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PAPER PRESENTATION AND PUBLICATION:

- **Two Factor Authentication Through Morse Code Using Eye Blinks** – **First author** to this paper, published in International Journal of All Research Education and Scientific Methods (**IJARESM**), in 2022. [Link](#).
 - Presented a seminar on the merits, demerits and plausible uses of deep learning methods on electric power system for fault diagnosis and presented a work around for increasing the efficiency of diagnosis.
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